Multisystem Inflammatory Syndrome in Children

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Introduction

- ☑ Multisystem inflammatory syndrome in children (MIS-C) associated with coronavirus disease-2019 (COVID-19) or pediatric inflammatory multisystem syndrome (PIMS) was initially reported in the United Kingdom and the United States in April 2020 following a surge in COVID-19 infections in the population.
- ☑ It has the presentation of hyperinflammatory syndrome with involvement of multiple organs, requiring timely treatment of anti-inflammatory drugs such as steroids and intravenous immunoglobulins.
- ☑ It is an uncommon but potentially serious disease in children and adolescents; timely diagnosis and treatment is associated with good outcomes.

Diagnostic Criteria

In May 2020, the World Health Organization (WHO) and the Centers for Disease Control (CDC) issued separate diagnostic criteria based on available/published information in case reports and case series analysis. These criteria are enlisted in **Table 1**.

Diagnostic Criteria

TABLE 1: The WHO criteria for MIS-C.

Criteria	***All 6 criteria must be met***
Age	Age 0–19 years
Fever	Fever for ≥3 days
Clinical signs of multisystem involvement (at least two of the following)	☑ Rash, bilateral nonpurulent conjunctivitis, or mucocutaneous inflammation signs (mouth, hands, or feet) ☑ Hypotension or shock cardiac dysfunction, pericarditis, valvulitis, or coronary abnormalities (including echocardiographic findings or elevated troponin/BNP) ☑ Evidence of coagulopathy (prolonged PT or PTT; elevated D-dimer) ☑ Acute gastrointestinal symptoms (diarrhea, vomiting, or abdominal pain)
Elevated markers of inflammation	Elevated markers of inflammation (e.g., ESR, CRP, or procalcitonin)
Rule out other diagnoses	No other obvious microbial cause of inflammation, including bacterial sepsis and staphylococcal/streptococcal toxic shock syndromes/tropical infectious diseases, i.e., malaria, dengue, scrub typhus, leptospirosis, and enteric fever
Recent or current SARS-CoV-2 infection or exposure	☑ Any of the following tests positive: – Positive SARS-CoV-2 RT-PCR – Positive serology – Positive antigen test – Contact with an individual with COVID-19

(BNP: B-type natriuretic peptide; CRP: C-reactive protein; COVID-19: coronavirus disease-2019; ESR: erythrocyte sedimentation rate; MIS-C: multisystem inflammatory syndrome in children; PT: prothrombin time; PTT: partial thromboplastin time; RT-PCR: reverse transcription–polymerase chain reaction; SARS-CoV-2: severe acute respiratory syndrome coronavirus-2; WHO: World Health Organization)

Investigations

Investigation in MIS-C depends upon severity of disease. Approach to a child with features suggestive of MIS-C:

- ☑ *Tier-1 tests*: Complete blood count (CBC), C-reactive protein (CRP), erythrocyte sedimentation rate (ESR), liver function test (LFT), kidney function test (KFT), blood sugar, blood gas, and severe acute respiratory syndrome coronavirus-2 (SARS-COV-2) serology/reverse transcription–polymerase chain reaction (RT-PCR).
- ☑ *Tier-2 tests*: Electrocardiogram (ECG), echocardiogram, B-type natriuretic peptide (BNP), troponin-T, ferritin, lactate dehydrogenase (LDH), procalcitonin, interleukin 6 (IL-6), prothrombin time (PT), activated partial thromboplastin time (aPTT), D-dimer, and fibrinogen.

Positive tier 1 screen (both of these should be present):

1. CRP > 5 mg/dL and/or ESR > 40 mm/hour
2. At least one of these: Absolute lymphocyte count (ALC) < 1,000/µL, platelet count < 150,000/µL, Na < 135 mEq/L, neutrophilia, and hypoalbuminemia

Isolated increased COVID-19 antibodies are *not synonymous* with MIS-C. For diagnosis of MIS-C, it is mandatory to rule out common tropical infections including malaria, dengue, enteric fever, rickettsial illness (scrub typhus), etc.

Multisystem inflammatory syndrome in children associated with COVID-19 can present as critical illness. There can be a spectrum of presentations from mild symptoms to multiorgan dysfunction syndrome.

Steps of Management

- ☑ Stabilize the patient (airway stabilization and adequate perfusion)
- ☑ Treat organ dysfunction and prevent further progression (beware of organ dysfunction)
- ☑ Control of systemic inflammation by choosing the right immunosuppression
- ☑ Close monitoring for disease progression
- ☑ Long-term follow-up for complications.

ABCD: **A**irway stabilization and **A**dequate perfusion, **B**eware of organ dysfunction, **C**ontrol systemic inflammation, and close monitoring for **D**isease progression

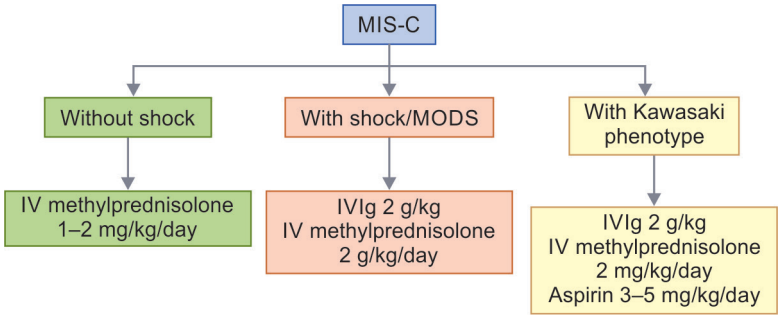
Clinical Types

Multisystem inflammatory syndrome in children for management purposes can be grouped into four categories **(Flowchart 1)**:

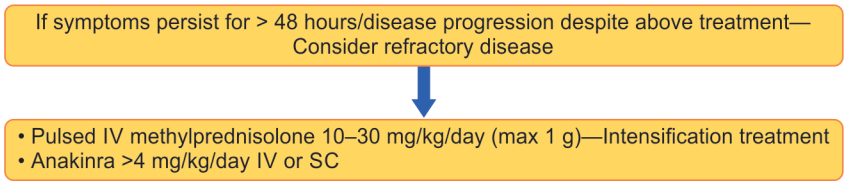
1. *MIS-C without shock*: Any child who fulfills the WHO criteria for MIS-C and stable without any feature of shock
2. *MIS-C with shock/multiple organ dysfunction syndrome (MODS)*: Any child who fulfills the WHO criteria for MIS-C and having features of shock in the form of tachycardia, hypotension, requiring fluid bolus ≥ 30 mL/kg, or inotropic support. *MODS*: Any child who fulfills the WHO criteria for MIS-C and has two or more organ involvement [respiratory/cardiac/central nervous system (CNS)/liver/renal]
3. *MIS-C with Kawasaki phenotype*: Children who meet complete or incomplete Kawasaki disease criteria as defined by the American Heart Association (Kawasaki diagnosis is established by fever lasting 5 or more days and at least four of the following five clinical criteria: Polymorphous rash (excluding bullous or vesicular eruptions); Conjunctival injection; Oropharyngeal mucous membrane changes; Extremity changes; and Lymphadenopathy)
4. *MIS-C with refractory disease*: Any child who fulfills the WHO criteria for MIS-C and has not responded to first tier therapy [intravenous immunoglobulin (IVIg) and low-dose steroids) after 48 hours.

Steps of Management

Flowchart 1: Management of MIS-C.



(IV: intravenous; IVIg: intravenous immunoglobulin; MIS-C: multisystem inflammatory syndrome in children; MODS: multiple organ dysfunction syndrome)



☑ *Airway stabilization and adequate perfusion*:

- These children should preferably be monitored in intensive care unit (ICU). Shock can be vasodilatory/cardiogenic.
- Judicious fluid resuscitation 10–20 mL/kg over 30–60 minutes and aggressive hemodynamic support with prompt initiation of vasoactive agents.
- Epinephrine can be used if there is hypotension with cardiac involvement, norepinephrine if there is vasodilatory shock aiming good mean arterial pressure for adequate organ perfusion.
- In extreme cases with catecholamine, refractory shock vasopressin is advised.

☑ *Treat organ dysfunction and prevent further progression (beware of organ dysfunction)*:

- Antibiotics in first hour after obtaining blood cultures as per local hospital antibiotic guidelines
- Prevent organ dysfunction by maintaining good organ perfusion
- Avoid fluid overload.

Steps of Management

- ☑ *Control of systemic inflammation by choosing the right immunosuppression*: This therapy is mainly targeted to reduce tissue inflammation or prevent progression of coronary artery aneurysm/myocardial dysfunction. Initial combined treatment with IVIg and corticosteroids may be beneficial.
  - *IVIg*: Dose: 2 g/kg (based on ideal body weight with maximum dose of 100 g) IV. This can be given as a single infusion over 8–12 hours or 12–24 hours based on patient's clinical status and cardiac function. In children who fail to respond, second dose may be considered.
  - *Methylprednisolone*: Should be administered simultaneously with IVIg at low dose of 2 mg/kg/day; however in children with coronary artery changes or refractory disease, pulsed dose of 10–30 mg/kg (maximum of 1,000 mg) may be administered. This is slowly transitioned to oral prednisolone which is tapered over 2–3 weeks with clinical and CRP monitoring.
  - *Anakinra*: In children with refractory disease despite glucocorticoid treatment or in patients with contraindications to steroids, anakinra at dose of >4 mg/kg/day IV or SC should be considered after expert consult.
  - *Infliximab/tocilizumab*: Currently not recommended for use in children.
- ☑ *Close monitoring and disease progression*:
  - Vigilant clinical monitoring for progression into hemodynamic instability or organ involvement should be done.
  - Laboratory monitoring of inflammatory markers is recommended till patient is stable.
  - ECG and echocardiogram have to be repeated after 48 hours as per clinician's discretion, subsequent echocardiogram at 1–2 weeks, 4–6 weeks, and 1 year if initial echocardiogram abnormal.
  - If child appears unwell or deteriorates after 24–48 hours of treatment, consider expert consult.
- ☑ *Anticoagulation*:
  - MIS-C with documented thrombosis/ejection fraction < 35%/coronary artery Z score ≥ 10/giant aneurysm with diameter > 8 mm: Enoxaparin 1 mg/kg (0.75 mg/kg/dose in <2 months) SC for 2 weeks after discharge.
  - Low-dose aspirin (3–5 mg/kg/day; maximum 80 mg/day) should be used if platelets >80,000/µL and continued till normal coronary arteries are confirmed at ≥4 weeks after diagnosis.

  In patients with aneurysm or risk of thrombosis, it is desirable to continue antiplatelet and anticoagulation as per their risk/need with clinicians' judgment.
- ☑ *Long-term follow-up for complications*: Mortality is reported in 1–2% of affected patients; higher figures are often reported with delayed presentations. Coronary artery aneurysm occurs in 25%, cardiac dysfunction in 50–60%, respiratory failure in 30%, renal involvement in 12%, CNS in 3%, and systemic thrombosis in 3–6%.

Further Reading

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